RACE Regular Analysis Through Continuous Exercise

Tangent and Normal : RACE-01

Single Correct Question

1. The side of a square sheet is increasing at the rate of 4 cm per minute. The rate by which the area increasing when the side is 8 cm long is-

(A) 16 $cm^2/min.$ (B) 32 $cm^2/min.$ (C) 64 $cm^2/min.$ (D) 80 $cm^2/min.$

MTN121

2. The radius of an air bubble is increasing at the rate of 0.5 cm/sec. The rate by which the volume of the bubble is increasing when the radius is 1 cm, is-

(A) $2\pi cm^3/s$ (B) $\pi\ cm^3/s$ (C) $4\pi cm^3/s$ (D) $0.5\pi cm^3/s$

MTN122

3. The point on the curve $y^2 = 8x$ for which the abscissa and ordinate change at the same rate is-

(A) $(4, 2)$ (B) $(-4, -2)$ (C) $(2, 4)$ (D) $(-2, -4)$

MTN123

4. A particle moves along the curve $y = x^2 + 2x$. Then the points on the curve where the x and y coordinates of the particle changing at the same rate, are

(A) $\left(\frac{-3}{4}, \frac{-1}{2}\right)$ (B) $\left(\frac{-1}{2}, \frac{-3}{4}\right)$ (C) $\left(\frac{3}{4}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \frac{3}{4}\right)$

MTN124

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5. Using differentials, find the approximate value of $\sqrt{25.2}$.

(A) 5.02 (B) 5.01 (C) 5.03 (D) 5.04

MTN125

6. The tangent to the curve $y = ax^2 + bx$ at $(2, -8)$ is parallel to x-axis. Then

(A) $a = 2, b = -2$ (B) $a = 2, b = -4$ (C) $a = 2, b = -8$ (D) $a = 4, b = -4$

MTN126

7. The point of the curve $y^2 = 2(x - 3)$ at which the normal is parallel to the line $y - 2x + 1 = 0$ is

(A) $(5, 2)$ (B) $\left(-\frac{1}{2}, -2\right)$ (C) $(5, -2)$ (D) $\left(\frac{3}{2}, 2\right)$

MTN127

8. At which point the line $\frac{x}{a} + \frac{y}{b} = 1$, touches the curve $y = be^{-x/a}$

(A) $(0, 0)$ (B) $(0, a)$ (C) $(0, b)$ (D) $(b, 0)$

MTN128

9. The area of the triangle formed by the coordinate axes and a tangent to the curve $xy = a^2$ at the point (x_1, y_1) on it is

(A) $\frac{a^2x_1}{y_1}$ (B) $\frac{a^2y_1}{x_1}$ (C) $2a^2$ (D) $4a^2$

MTN129

10. An equation of the tangent to the curve $y = x^4$ from the point $(2, 0)$ not on the curve is

(A) $y = 0$ (B) $x = 0$ (C) $x + y = 0$ (D) None of these

MTN130

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11. The sum of intercepts on co-ordinate axes made by tangent to the curve $\sqrt{x}+\sqrt{y}=\sqrt{a}$ is

(A) a (B) $2a$ (C) $2\sqrt{a}$ (D) None of these

MTN131

Multiple Correct Question

12. If tangent to curve $2y^3 = ax^2 + x^3$ at point (a, a) cuts off intercepts α, β on co-ordinate axes, where $\alpha^2 + \beta^2 = 61$, then the value of $'a'$ is equal to

(A) 20 (B) 25 (C) 30 (D) –30

MTN132

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer	C	A	C	B	A	C	C	C	C	A	A	C,D

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RACE Regular Analysis Through Continuous Exercise

Tangent and Normal : RACE-02

Single Correct Question

1. The slope of tangent to the curve $x = t^2 + 3t - 8, y = 2t^2 - 2t - 5$ at the point $(2, -1)$ is

(A) $\frac{22}{7}$ (B) $\frac{6}{7}$ (C) -6 (D) None of these

MTN133

2. If $x = t^2$ and $y = 2t$, then equation of the normal at $t = 1$ is

(A) $x + y - 3 = 0$ (B) $x + y - 1 = 0$ (C) $x + y + 1 = 0$ (D) $x + y + 3 = 0$

MTN134

3. The normal of the curve $x = a(\cos\theta + \theta\sin\theta), y = a(\sin\theta - \theta\cos\theta), (a \neq 0)$ at any θ is such that

(A) It makes a constant angle with x-axis

(B) It passes through the origin

(C) It is at a constant distance from the origin

(D) None of these

MTN135

4. The normal to the curve $x = a\,(1 + \cos\theta), y = a\sin\theta$ at 'θ' always passes through the fixed point

(A) (a, a) (B) $(0, a)$ (C) $(0, 0)$ (D) $(a, 0)$

MTN136

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5. The equation of the tangent to the curve $x = 2\cos^3\theta$ and $y = 3\sin^3\theta$ at the point $\theta = \pi/4$ is

(A) $2x + 3y = 3\sqrt{2}$ (B) $2x - 3y = 3\sqrt{2}$ (C) $3x + 2y = 3\sqrt{2}$ (D) $3x - 2y = 3\sqrt{2}$

MTN137

6. If the tangent to the curve $x = a\,(\theta + \sin\theta), y = a(1 + \cos\theta)$ at $\theta = \frac{\pi}{3}$ makes an angle α $(0 \leq \alpha < \pi)$ with x-axis, then $\alpha =$

(A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{5\pi}{6}$

MTN138

7. If the normal at the point $(3t, 4/t)$ of the curve $xy = 12$ cuts the curve again at $(3t_1, 4/t_1)$ then find 't_1' in terms of 't'

(A) $\frac{-9}{16t^3}$ (B) $\frac{-16}{9t^3}$ (C) $\frac{9}{16t^3}$ (D) $\frac{16}{9t^3}$

MTN139

8. The angle of intersection of the curves $y = x^2$ and $x = y^2$ at $(1, 1)$ is

(A) $\tan^{-1}\left(\frac{4}{3}\right)$ (B) $\tan^{-1}(1)$ (C) 90° (D) $\tan^{-1}\left(\frac{3}{4}\right)$

MTN140

9. If the curve $y = a^x$ and $y = b^x$ intersect at angle α then,$\tan\alpha =$

(A) $\frac{a-b}{1+ab}$ (B) $\frac{\log a - \log b}{1+\log a \log b}$ (C) $\frac{a+b}{1-ab}$ (D) $\frac{\log a + \log b}{1-\log a \log b}$

MTN141

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10. For the curve $xy = c^2$ the subnormal at any point varies as

(A) x^2 (B) x^3 (C) y^2 (D) y^3

MTN142

11. For the curve $by^2 = (x + a)^3$ the square of subtangent is proportional to

(A) $(\text{Subnormal})^{1/2}$ (B) Subnormal

(C) $(\text{Subnormal})^{3/2}$ (D) None of these

MTN143

12. The subtangent, ordinate and subnormal to the parabola $y^2 = 4ax$ at a point (different from the origin) are in

(A) AP (B) GP (C) HP (D) AGP

MTN144

Multiple Correct Question

13. If at the point $\theta = \frac{\pi}{2}$ on the curve $x = a(\theta + \sin\theta), y = a(1 - \cos\theta), a \neq 1$, then

(A) length of normal = $\sqrt{2}a$ (B) Length of sub-tangent = a

(C) length of sub-normal = a (D) Length of tangent = $\sqrt{2}a$

MTN145

Matrix Match Type Question

14.

	Column – I		Column – II
(A)	If curves $y^2 = 4ax$ and $y = e^{-\frac{x}{2a}}$ are orthogonal then 'a' can take value	(p)	3
(B)	If θ is angle between the curves $y = [\lvert\sin x\rvert + \lvert\cos x\rvert]$, ($[\cdot]$ denote GIF) and $x^2 + y^2 = 5$ then $\text{cosec}^2\theta$ is	(q)	1
(C)	If curves $y^2 = 4a\,(x + a)$ and $y^2 = 4b(x + b)$ intersects each other orthogonally then $\left\lvert\frac{a}{b}\right\rvert$ can be equal to___	(r)	5/4
(D)	If $y = x^2 + 3x + c$ and $x = y^2 + 3y + c$ touches each other at (h, k) then $\lvert h + k + c\rvert$ can be equal to...	(s)	2

MTN146

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	C	D	C	D	B	D	B	D
Question	11	12	13	14						
Answer	B	B	A,B,C,D	A → p,q,r,s; B → r; C → p,q,r,s; D → q						

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